

Language Model Benchmark Contamination Leaves a Person-Fit Signature in the Graded Response Matrix

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Abstract

A benchmark score is trustworthy only if the evaluated model has not trained on the test items, yet existing contamination detectors demand privileged access — model weights, token log-probabilities, or the training corpus — that a public leaderboard entry does not expose. We propose a detector requiring only the graded response matrix: which items each model answered correctly. Treating contamination as psychometric item preknowledge, a contaminated model behaves like an aberrant test-taker, succeeding on items too difficult for its true ability because it has memorized them. Under controlled fine-tuning injection across four base models and three benchmarks, contamination separates from both domain-matched and generic benign fine-tuning at AUROC 0.94–0.99, and the separation survives when memorization is isolated from collateral training damage. The signal is semantic, not verbatim: it survives paraphrasing of evaluation items, where accuracy-based detectors fail. A signed decomposition of the aberrance extends to the mirror problem of sandbagging, placing the two failure modes at opposite tails of a single ability–response axis. On released models, the detector yields near-zero flags on contamination-resistant benchmarks and confirms documented positives, agreeing with a log-probability detector that requires strictly more access. Person-fit flags aberrance, of which contamination is one cause; near-total memorization is absorbed into estimated ability and invisible. We therefore position the method as an access-agnostic triage and flagging tool, not a standalone classifier.

Introduction

Public leaderboards rank language models by benchmark accuracy, and those rankings increasingly drive research direction, procurement, and deployment. The rankings are only meaningful if evaluated models have not seen the test set during training. Yet benchmark contamination, evaluation items leaking into training data, is pervasive and hard to rule out (Sainz et al. 2023), especially for models released as weights or as API endpoints with undisclosed training corpora.

Most existing contamination detectors require privileged access: membership inference from token log-probabilities needs the model’s weights or logits, and n -gram overlap needs its training corpus. A typical leaderboard entry exposes neither, recording only which items each model answered correctly. We ask whether contamination can be detected from that graded response matrix alone.

We apply a classical idea from psychometrics: person-fit statistics flag a response vector as aberrant when its pattern of correct and incorrect answers is improbable, given the test-taker’s ability. We hypothesize that a contaminated model will succeed on memorized items that are “difficult” given its true ability; this framework connects contamination to the reverse failure, sandbagging, in which a model deliberately fails on items that are “easy” given its true ability.

Both failure modes give rise to deployment concerns: a misaligned model under evaluation for release may selectively underperform on items that pre-deployment tests intend to flag; conversely, a contaminated model trained on a safety refusal benchmark appears to exhibit stronger safeguards than it has. In both cases the benchmark result no longer reflects the properties it is trusted to evaluate, leaving the underlying hazardous capabilities unmitigated.

We demonstrate the contamination-versus-sandbagging unification on a hazardous-knowledge benchmark (WMDP; Li et al. 2024), using the fine-tuned sandbagging Model Organisms released by the UK AI Security Institute (Bloom et al. 2025; UK AI Security Institute (aisi-whitebox) 2025) against contaminated models on the same items.

Contributions. We make the following contributions.

- A contamination detector operating on the graded response matrix alone, validated causally under controlled fine-tuning injection with a dose-response relationship and two classes of benign fine-tuning control—including the finding that naive person-fit on multiple-choice items is spuriously perfect unless the standard guessing floor is modeled.
- Evidence that the signal is semantic, not verbatim: it survives paraphrasing of the evaluation items, where accuracy-based detectors fail.
- A signed decomposition of person-fit misfit that places contamination and sandbagging on opposite tails of a single ability–response construct, with initial small-sample evidence that one signed axis separates them.

Related Work

Contamination detection. We organize existing contamination detectors by their required level of access. *Training-corpus access* methods (e.g. n -gram overlap) compare evaluation items against the training text directly, an upper bound

available only to model developers. *Weight or log-probability access* methods, including Min-K% (Shi et al. 2023) and its distribution-normalized successor Min-K%++ (Zhang et al. 2024), perform membership inference from per-token likelihoods, and the exchangeability permutation test of Oren et al. (2024) requires the model’s per-example log-probabilities over the benchmark text. A *behavioral* tier needs neither corpus nor weights but re-prompts the model and inspects its generated text: guided-instruction completion (Golchin and Surdeanu 2024), the multiple-choice Data Contamination Quiz (Golchin and Surdeanu 2025), test-set-slot guessing (Deng et al. 2024), and output-distribution peakedness (Dong et al. 2024) detect memorization from model outputs, though each generates fresh outputs and most are keyed to a specific probe (a masked option, a perturbed variant, a completion prefix). Our *response-matrix-only* method instead uses only per-item correctness, the level at which archived leaderboards publish.

IRT for LLM evaluation. A concurrent line applies item-response theory to language-model evaluation. DualEval (Li et al. 2026) defines signed per-(model,item) residuals and recovers synthetically injected contamination at high cell-level AUROC against a clean-fit reference, but operates per cell, yields no per-model aberrance statistic, and does not treat sandbagging. Land and Bikel (2026) audit benchmark *items* for mislabeling with a four-parameter IRT fit; the single model they flag as contamination-consistent is an incidental screening signal from its agreement with those label-errors, not detection via item preknowledge. Person-side aberrance is itself long-established in psychometrics (Meijer and Sijsma 2001; Karabatsos 2003; Sijsma 1986; van der Flier 1980, 1982). Strugatski and Alexandron (2025) use person-fit statistics to distinguish human test-takers from those using an LLM to cheat. Concurrent work under review (Anonymous 2025) applies off-the-shelf plain-Rasch person-fit statistics to the mirror problem of sandbagging alone, detecting fine-tuned and prompted underperformance on the AISI model organisms as a triage signal; it does not treat contamination, guessing-aware correction, or the signed decomposition that unifies the two failure modes. From the same IRT residuals, we instead construct a single per-model, signed statistic spanning contamination and its mirror image, sandbagging, computable from one response row, using only pre-calibrated item difficulties as a reference.

Methods

We observe a binary response matrix $X \in \{0, 1\}^{N \times J}$: for N models and J items, $X_{ij} = 1$ iff model i answered item j correctly (missing entries are masked). We calibrate an item model on a fixed *baseline cohort* of $N = 17$ models to estimate per-item difficulties b_j . Then, for model i , we estimate its ability θ_i against the baseline-computed b_j .

Item difficulty. Our item model is the guessing-aware Rasch model (IPL-G),

$$P(X_{ij} = 1) = g_j + (1 - g_j) \sigma(\theta_i - b_j), \quad (1)$$

with the guessing floor $g_j = 1/n_j$ fixed a priori to the reciprocal of item j ’s number of choices ($g = 0$ for free-form

items). This captures the nonzero probability of a correct guess on multiple-choice items that a plain Rasch model (Birnbaum 1968) ignores. We fix g and omit a free per-item discrimination rather than estimate them: both are poorly determined and unstable to estimate at our cohort size, with discrimination prone to degenerate runaway on near-separated items (Lord 1980). Difficulties b are fit once on the baseline cohort by maximum likelihood over the response matrix and then frozen¹.

Model ability. We estimate a test model’s ability $\hat{\theta}$ by maximum likelihood under the baseline b (an in-sample MLE on that model’s own response row). The Snijders correction accounts for $\hat{\theta}$ being estimated rather than known. It does not account for a second problem: a contaminated row biases its own $\hat{\theta}$, because the memorized successes that inflate the ability estimate are also the aberrance being scored. We address this in-sample bias through the ability estimator, analyzing two variants that break the circularity: an oracle estimator restricted to manifest-known untrained items, and a deployable robust estimator. The choice affects the signed attribution but not detection (the guessing-aware Z^+ detection AUROC is nearly identical in-sample and under the oracle). The oracle variant is reported only as a controlled-setting upper bound.

Person-fit statistics. Person-fit statistics are a mature psychometric family for flagging response vectors that are improbable given a fitted ability (Meijer and Sijsma 2001; Karabatsos 2003). Our primary statistic is the standardized log-likelihood person-fit statistic l_z^* with the finite-cohort correction of Snijders (2001), which accounts for θ being estimated rather than known. As robustness comparators we report covariance-based H^T (Sijsma 1986; Sijsma and Meijer 1992), and Guttman-error-based U_3 (van der Flier 1980, 1982; Guttman 1944). We connect our signed decomposition to the item-preknowledge literature (Sinharay 2017; van Krimpen-Stoop and Meijer 2001).

Empirical Null and Robust z Person-fit statistics have no absolute scale, so we always report a statistic relative to a null built from the cohort itself. Leaving one cohort model out at a time (LOMO), we refit the item difficulties b on the remaining models and re-score every model under that refit; the resulting values are an empirical null for how aberrant a clean model looks. We summarize each scored model’s position as a robust z (z_{rob}): its distance above the null median in MAD units. We use the MAD rather than the standard deviation because the null is small and heavy-tailed.

Wherever ground-truth labels exist—the controlled injection experiments—our primary evidence is the *separation* between conditions, reported as an AUROC, and no threshold is involved. A threshold is needed only to *count* how often the detector fires on real released models, for which no contamination labels exist (see Applying the Detector to Released Models). We use a single flag threshold at $z_{\text{rob}} > 3$

¹The calibration is stable to cohort composition despite the modest N : the contamination-vs-benign separability verdict is invariant across leave- k -out resamples (worst case, leave-three-out on ARC, AUROC 0.88; Supplement S4).

as a reporting instrument, not a classifier verdict. Because the null is heavy-tailed, $z_{\text{rob}} > 3$ does not imply the nominal Gaussian 0.1% false-positive rate. We instead measure the rate empirically—the fraction of clean cohort models crossing the line—and report it with every flag count. Supplement S2 gives the exact scaling and confirms the flag counts are insensitive to null and estimator conventions.

A Guessing-Aware, Signed Decomposition We split misfit into a positive channel Z^+ (unexpected successes, the contamination signature) and a negative channel Z^- (unexpected failures, the sandbagging signature), and take their difference as a signed discriminant $d_{\text{norm}} = Z^+ - Z^-$. Each channel is a z -score of weighted unexpected successes (Z^+) or failures (Z^-), centered and scaled to mean 0, variance 1 under the item model at $\hat{\theta}$ (exact form in the supplement).

Contamination detection uses only the positive channel Z^+ , whose contamination-vs-clean separation is, empirically, essentially unchanged whether or not the guessing floor is modeled. Attribution is different: it uses the signed difference d_{norm} and so depends on the failure channel Z^- , for which the guessing floor has opposing consequences. A plain Rasch fit reports a spuriously perfect separation (an artifact; see Contamination and Sandbagging as One Construct), but modeling it makes Z^- fragile. The failure channel weights each item by how surprising a failure would be, and for a very easy item (a failure would be very surprising) that weight grows without bound. Such items are normally answered correctly and contribute nothing, but a single one flipped to incorrect by fine-tuning damage can then dominate Z^- . We therefore *winsorize*: we cap each item’s success probability at an upper level ϕ , which bounds the failure weight on the cohort-unanimous items (answered correctly by all models) that carry no aberrance signal anyway. We fix $\phi = 0.99$ in advance from the cohort rather than tuning it to the outcome: with our 17 calibration models about 27% of items are cohort-unanimous, and every ϕ across the plateau ($\phi \geq 0.97$) clips exactly this same empirically-unanimous set. The resulting separation does vary with ϕ , so we report it as a band over the plateau rather than at a single clip; clipping more aggressively than the plateau does not weaken it, so the reported band is the conservative end of the range. The clip touches both channels but materially affects only Z^- .

Evaluation protocol. Every reported AUROC is the rank-based area under the ROC curve, with stratified percentile bootstrap 95% CIs (Efron and Tibshirani 1993). *Detection* AUROC scores each condition’s statistic (positive) against the LOMO baseline null (negative). *Separability* AUROC scores contaminated conditions (positive) against a benign negative set: the domain-matched and generic fine-tuning controls for the specificity claim, or the sandbagging conditions for the attribution claim. Unless stated otherwise, separability restricts contamination to injected rates of 25% and above (the fully memorized rate is excluded, being invisible by construction; see Limitations and Scope). Higher statistic values denote more aberrance throughout.

We validate the detector in four settings, presented in turn: controlled injection, paraphrase robustness, the contamination-sandbagging construct, and released models in the wild.

Detecting Injected Contamination Under Known Ground Truth

To validate the detector under known ground truth, we inject contamination by fine-tuning: we use a publicly released base model, fine-tune it on a controlled fraction of the evaluation items, and measure whether person-fit recovers the manipulation. The grid spans four base models and three benchmarks at contamination rates from a few percent to full memorization, each at two random seeds.

Detection and dose-response. Person-fit misfit tracks contamination rate (Figure 1): even at the lowest injected rates the statistic already sits many standard deviations above the calibrated null (the detection signal), and it broadly increases with rate. To disambiguate fine-tuning effects (which also perturb response patterns), we measure condition separation from benign fine-tuning: both a domain-matched control (fine-tuned on held-out data from the same domain) and a generic instruction-tuning control perturb the response pattern and register elevated person-fit. We demonstrate detector specificity: contamination separates at AUROC 0.94–0.99 on the multiple-choice benchmarks and near-perfectly on the free-form benchmark (nominal-vs-effective dose caveat in Supplement S0). This signal holds across models and benchmarks: Table 1 reports the per-cell guessing-aware Z^+ separability AUROC (contaminated conditions vs. the two benign fine-tuning controls) for every base–benchmark cell we ran, alongside the median contaminated Z^+ effect size. At contamination rates of 25% and above, contaminated models separate from both benign controls at (per-base) AUROC 1.00 on all three benchmarks on the l_z^* statistic, and equally on the guessing-aware Z^+ channel (pooled numbers in the Table 1 caption).

We can isolate the training damage confound. Aggressive memorization fine-tuning also damages performance on untrained items, which itself perturbs person-fit. The injected condition is therefore a compound signature (memorization plus collateral damage) that upper-bounds a real leak. A synthetic clean-leakage construction isolates the two components: we take each base model’s own clean response row, set injected items to correct (simulate memorization) with no collateral flips, and score it under the identical IPL-G calibration. Pooled across all four base models, this realistic damage-free contamination still separates from benign fine-tuning at guessing-aware Z^+ AUROC 1.00 (MMLU) and 0.99 (ARC, 95% CI [0.95, 1.00]). The realistic effect size is modestly below the damaged one (median Z^+ 10.7 vs. 12.3 on MMLU) rather than being created by the damage. Because this construction involves no fine-tuning at all, neither collateral damage nor training intensity can explain the separation (Supplement S1).

Localization. Beyond flagging a model, per-item suspicion scores identify which specific items were contaminated. Against the effective contaminated set, paired suspicion reaches median precision@ k of 0.653 at a prevalence of 0.07 (localization AUROC 0.975, bootstrap 95% CI [0.97, 0.98] over conditions); a deployable single-model variant reaches precision@ k 0.529 and localization AUROC

Benchmark	Base model	n_c/n_b	med. Z^+	AUROC
MMLU (MCQ)	Llama-3.1-8B	8/2	12.7	1.00
	Qwen2.5-7B	4/2	12.0	1.00
	Mistral-7B-v0.3	4/2	13.6	1.00
	OLMo-3-7B	4/2	10.9	1.00
GSM8K (free)	Llama-3.1-8B	8/2	21.7	1.00
	Qwen2.5-7B	4/2	18.2	1.00
ARC (MCQ)	Llama-3.1-8B	8/2	5.2	0.88
	Qwen2.5-7B	4/2	4.5	1.00
	Mistral-7B-v0.3	4/2	7.3	1.00
	OLMo-3-7B	4/2	5.4	1.00

Table 1: Controlled-contamination grid. For each base–benchmark cell, the guessing-aware Z^+ separability AUROC separates all contaminated conditions (across injected rates, two seeds) from the two benign fine-tuning controls (domain-matched + generic); n_c/n_b is the count of contaminated vs. benign conditions and “med. Z^+ ” the median contaminated misfit ($\geq 25\%$ rate). Multiple-choice benchmarks MMLU (Hendrycks et al. 2021) and ARC (Clark et al. 2018) are computed under the guessing-aware 1PL-G floor ($g = 1/4$) with the in-sample $\hat{\theta}$ (Supplement S0); GSM8K (Cobbe et al. 2021) is free-form ($g = 0$). Llama×ARC has AUROC 0.88, driven by the lowest injected rates (5–10%, where Z^+ overlaps the controls); at $\geq 25\%$ that cell too separates perfectly (1.00). Pooling the $\geq 25\%$ contaminated conditions against all benign controls per benchmark gives separability AUROC 1.00 on all three (MMLU/GSM8K/ARC), stratified percentile bootstrap 95% CI [1.00, 1.00] (Supplement S0). The lower headline band (0.94–0.99) instead pools every contamination rate across base models, whose Z^+ scores are not on a common scale and so rank-invert on a few of the lowest-dose conditions; restricted to a single base model or to rates $\geq 25\%$, separability returns to ≈ 1.00 . Cells have small n ($n_c \leq 8$); CIs are reported throughout the paper for this reason.

0.902 (CI [0.89, 0.91]), and benign controls sit at prevalence. Unlike a cell-level residual detector, which needs a clean-fit reference model to score anomalous cells (Li et al. 2026), this variant localizes from a single model’s own response row, using no reference beyond the cohort item difficulties.

The Signal Is Semantic, Not Verbatim

Yang et al. (2023) show that simple rephrasing or translation of benchmark items evades n -gram-based decontamination filters and still lets a model overfit the benchmark. We test whether person-fit detects this paraphrase-surviving memorization. To this end, we take the contaminated adapters trained on the original evaluation items and re-score them on paraphrased versions of those items: the question stems are rewritten while the answer keys are held fixed. No new training is performed, so any surviving signal reflects semantic, answer-mapping memorization rather than string reproduction.

The person-fit aberrance survives paraphrasing (Figure 2). On MMLU the paraphrased person-fit statistic closely tracks

the verbatim one at every contamination rate, and contamination still separates from benign fine-tuning at AUROC 1.000 (excluding the fully memorized rate). This holds across base models: re-scoring each of four base models’ contaminated adapters (Llama-3.1-8B (Llama Team, AI @ Meta 2024), Qwen2.5-7B (Qwen Team 2024), Mistral-7B (Jiang et al. 2023), OLMo-3-7B (Team OLMo 2025)) on the paraphrased items, the guessing-aware Z^+ separates $\geq 25\%$ contamination from the benign controls in all conditions, with median contaminated Z^+ essentially unchanged from verbatim to paraphrase (e.g. +12.0 $\rightarrow$ +12.6 Qwen, +13.6 $\rightarrow$ +13.6 Mistral).

Pooled across bases (16 contaminated vs. 8 benign conditions), paraphrase separability is AUROC 1.00; per-base counts are too small for a meaningful CI (Supplement S0). On GSM8K the signal attenuates modestly, but every contaminated condition still flags far above the null (robust- z above twelve).

Accuracy-based detection fails on paraphrase. Contaminated accuracy transfers to the paraphrases: the paired original-versus-paraphrase accuracy gap is near zero and not significant at realistic contamination rates (McNemar’s test (McNemar 1947), $p \gg 0.05$), so any detector keyed to an accuracy drop fails to detect, while person-fit stays at +15 to +27. This both confirms the memorization is semantic and provides a controlled instance of the access-regime argument: the correctness pattern carries contamination signal that the correctness rate alone does not.

Contamination and Sandbagging as One Construct

Under an ability–response model contamination and sandbagging are related: a contaminated model succeeds on items that are hard given its estimated ability (unexpected success), while a sandbagging model fails items that are easy given its ability (unexpected failure). Both inflate person-fit misfit; we distinguish between them by decomposing misfit into a positive channel Z^+ (unexpected successes) and a negative channel Z^- (unexpected failures). Contamination is Z^+ -dominant and sandbagging Z^- -dominant, so the two occupy opposite sides of the $Z^+ = Z^-$ diagonal (Figure 3). The signed discriminant $d_{\text{norm}} = Z^+ - Z^-$ is the axis on which they separate (Supplement S7).

We report the attribution axis as a construct-level result (see Limitations and Scope). The detection channel Z^+ is robust and effectively item-model-invariant: contamination separates from benign fine-tuning at AUROC 0.982 under both the plain and guessing-aware item models. The two-sided axis that separates contamination from sandbagging is more delicate. Under a plain Rasch model it appears to separate perfectly, but this is a guessing-floor artifact: once the guessing floor is modeled, roughly a quarter of items sit at near-certain success for a median-ability model, and the unbounded failure-channel weight makes the uncorrected two-sided statistic collapse to chance, a cautionary result for the broader use of item-response models in language-model evaluation. Winsorizing the degenerate tail restores separation to a bounded AUROC of 0.86–0.91 (bootstrap

Contamination dose-response: aberrance increases with injection rate; benign-FT controls flag but separate

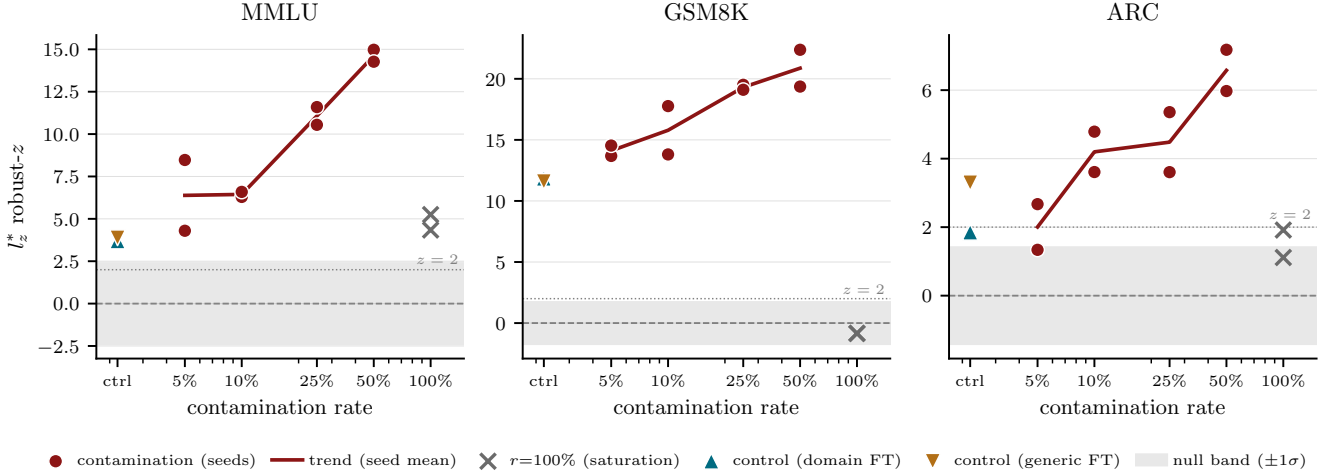


Figure 1: Contamination dose-response for Llama-3.1-8B, the base model run at five contamination rates (5–100%; the other three bases use two rates, Table 1). Person-fit misfit (robust z of l_z^* against the calibrated cohort null, shaded band) rises with contamination rate. We plot l_z^* for the dose-response view; the reported separability AUROCs use the guessing-aware Z^+ channel (Table 1). Domain-matched and generic fine-tuning controls (triangles) are elevated above the null, since any fine-tuning perturbs the pattern, but separate from contamination at rates of 25% and above. The fully memorized condition (gray ×) is invisible by construction on the free-form benchmark (see Limitations and Scope).

95% CI [0.76, 1.00]; Supplement S7); we report this band across the clip plateau ($\phi \geq 0.97$).

The estimate is small-sample by construction: $n = 6$ contaminated conditions against $n = 21$ sandbagging conditions drawn from only two sandbagging paradigms: a selective-prompt manipulation (van der Weij et al. 2024) applied across the calibration cohort, and the AISI model organisms fine-tuned to sandbag (Bloom et al. 2025; UK AI Security Institute (aisi-whitebox) 2025), one per malicious-mix rate. Two features of the design nevertheless argue that the separation is not a single-configuration artifact. First, the two sandbagging paradigms are induced by different routes (prompting, fine-tuning), while contamination can only be installed by training; the asymmetry is intrinsic to the failure modes, and because Z^- separates both routes from contamination, the signature tracks the underperformance itself rather than how it was installed. Second, the reported separation is a band over the a-priori winsorization plateau (A Guessing-Aware, Signed Decomposition). The AISI MO weights are no longer publicly hosted; because the detector consumes only the graded response matrix, we release the cached correctness vectors, and the result remains reproducible.

Applying the Detector to Released Models

We map where the detector applies among publicly released models, and where it does not (see Limitations and Scope for how these observations weigh against the controlled results). No ground-truth contamination labels exist for released models, so we instead report a specificity study on contamination-

resistant benchmarks (with an independent-cohort replication), a small set of externally documented positives, and a cross-check against a log-probability detector.

Specificity on contamination-resistant benchmarks. We calibrate a cohort item model on archived leaderboard responses and score 361 distinct models on GPQA (Rein et al. 2024) and MMLU-Pro (Wang et al. 2024), two benchmarks engineered to resist contamination. Applying the ability floor implied by 1PL-G, we obtain one above-floor flag across all three benchmark splits (GPQA-main 0/45, GPQA-Diamond 0/44, MMLU-Pro 1/286)². The corresponding empirical false-positive rate of the $z_{\text{rob}} > 3$ rule, measured by leave-each-out on the cohort itself rather than assumed from a Gaussian tail, is 0.00% (GPQA) and 0.36% (MMLU-Pro) among above-floor models: the counts here are rates of the rule as applied. Before the floor, the raw-sample rate is 2.5–13.6%: exactly the guessing-floor tail of near-chance models whose depressed ability inflates the floor residual, which the ability floor exists to remove. The single surviving flag is a near-chance model at the floor boundary, a residual instance of the same artifact, not a detection. A clean null on benchmarks engineered to resist contamination is the correct outcome: the method does not manufacture positives at scale. The null replicates on an independent, disjoint cohort—66 GPQA-Diamond models from a different era, separately calibrated, with zero above-floor flags (Supplement S5).

²A model can clear the ability floor on more than one split, so the per-split counts overlap and need not sum to 361.

Semantic, not verbatim: paraphrased items still flag
(MMLU unattenuated; GSM8K attenuates ~20–30% but stays far above null)

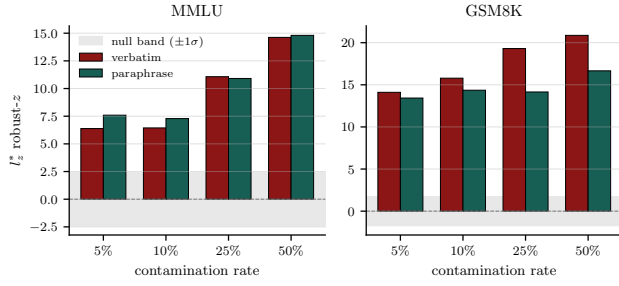


Figure 2: The signal is semantic, not verbatim. Person-fit misfit (robust z of l_z^*) for contaminated models scored on the original (verbatim) versus paraphrased evaluation items, by contamination rate. On the multiple-choice benchmark the paraphrased signal matches the verbatim one; on the free-form benchmark it attenuates modestly but remains far above the null band.

A documented-positive case study. In an earlier leaderboard era with externally documented contamination, and under the per-item guessing floor, we examine three distinct weight-released models with community-reported test-set inclusion. In a confirmatory check on these pre-flagged cases, each clears a robust- z of three on the contamination channel and separates from ability-matched controls at AUROC 0.949 (guessing-aware Z^+ ; 0.918 under plain Rasch).

Access-regime cross-check. We run Min-K%++, a log-probability membership-inference detector that requires model weights, on the weight-available positives: two of the three wild models (the third has had its weights withdrawn from public hosting) plus the synthetic positive control, for which weights are available by construction. The two detectors agree at the ranking level (pooled AUROC 0.970, 95% CI [0.88, 1.00], across 22 matched controls), but the person-fit signal fires far more strongly (robust- z up to an order of magnitude larger; Table 2) at strictly less access. This is some evidence for rank-level agreement across access regimes.

Limitations and Scope

Scope of claims. Our claims differ in evidential strength; we state the ordering once here rather than repeating it at each result. Strongest are the detection results, validated under known ground truth: a causal dose–response, two classes of benign fine-tuning control, a damage-free isolation of the memorization signal, and an operating envelope mapped over memorization breadth and selectivity. Two bounds apply. Installing contamination by fine-tuning is necessary for ground-truth labels but more targeted than the incidental, diffuse overlap of pretraining-scale contamination; the selectivity sweep shows detection extends to non-targeted uniform partial memorization, but diffuse pretraining-scale overlap is not directly tested and remains the main external-validity gap. And the clean multi-base paraphrase separation holds only

Contamination vs. sandbagging: winsorized attribution plane
(WMDP-Bio, 1PL-G guessing floor, $p \leq 0.99$)
AUROC (on d_{norm} , distance from boundary) = 0.90 [0.76, 1.00]

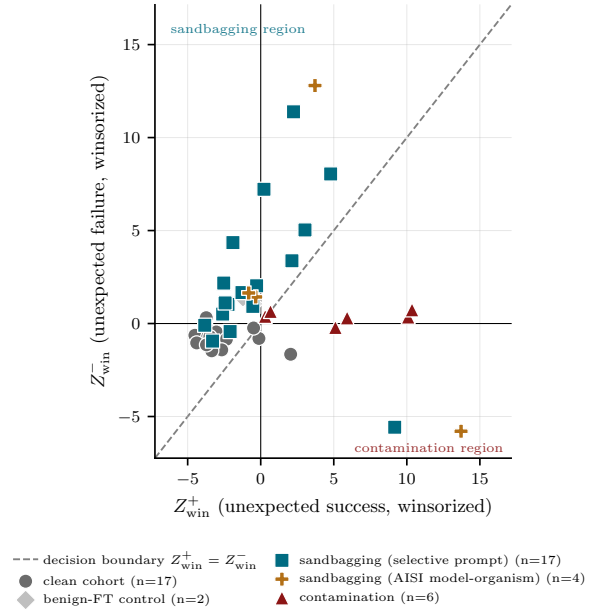


Figure 3: Contamination and sandbagging separate on the ratio of unexpected successes to failures. Both behaviors raise person-fit misfit, so both separate from the clean cohort. They differ in direction relative to the $Z^+ = Z^-$ diagonal (dashed): contaminated models fall below it (excess unexpected successes), sandbagging families above it (excess unexpected failures). Values are guessing-aware and winsorized; separation on the signed discriminant d_{norm} is AUROC 0.90 [0.76, 1.00].

in the multiple-choice setting; the free-form arm is single-base, scoped to targeted eval-set tuning above ecosystem background, and softens under paraphrase (Supplement S0).

Two further claims are weaker by design. The contamination-versus-sandbagging axis is a construct-level result: its confidence interval is wide, and because the two sandbagging paradigms are the independent units, even that interval understates uncertainty. It is initial evidence of separability, not a validated separator. The in-the-wild results are a scope map, not a leaderboard audit: no ground-truth contamination labels exist for released models, so the specificity study, independent-cohort replication, documented positives, and log-probability cross-check are converging observations rather than a powered detection rate. Scale compounds this: controlled injection is validated at 7–8B (the four base models), while the wild models range up to 70B; detectability at that scale is left to future work.

Failure to detect in a characterized corner. The method detects selective aberrance, in which there is a mismatch between empirical ability and item-level pattern. Uniform memorization is absorbed into estimated ability and remains invisible: we map this operational boundary directly with a graduated semi-synthetic sweep. On a clean base row we

Access tier	Signal required	Detector
Training corpus	item vs. train text	n -gram overlap
Weights / logits	per-token logprobs	Min-K%++, exch. test
Response matrix	per-item correct.	person-fit (ours)
<i>Per-positive signal (robust-z), weight-available cases:</i>		
Model	PF Z^+ (ours)	Min-K%++
Synthetic proof (7B) [†]	+14.2	+1.0
COKAL-v1 (70B)	+10.3	+1.3
una-xabarius (34B)	+3.1	+1.0

Table 2: Access-regime hierarchy and cross-check. Top: contamination detectors ordered by required access; person-fit sits at the least-privileged tier. Bottom: per-positive robust- z for the weight-available models. Person-fit fires sharply (up to +14) where Min-K%++, requiring strictly more access, sits near +1; the two agree on ranking (pooled AUROC 0.970). [†]Synthetic positive we fine-tuned, weights available by construction; the other rows are wild positives, a third having had its weights withdrawn.

memorize a fraction ρ of items along a selectivity knob α that interpolates from uniform-random ($\alpha = 0$) to hardest-items-first ($\alpha = 1$), re-estimate ability, and score the guessing-aware Z^+ against the cohort null (Figure 4). We make two observations. First, the detector fires ($z_{\text{rob}} > 3$) across most of the plane, including the uniform column: uniform partial memorization is detectable from $\rho \approx 0.15$ up to $\rho \approx 0.75$, because partial uniform leakage still perturbs the response pattern relative to ability. Near-total memorization ($\rho \rightarrow 1$) is invisible, where the row becomes (nearly) all-correct, ability runs away, and no per-item anomaly survives; we confirm the signal vanishes (in fact goes slightly negative) at full injection. This corner is the extreme of the in-sample circularity discussed above (inflated ability absorbs the memorized successes), not a separate failure mode. Second, an interior region where the deployable in-sample estimator collapses is an artifact of ability circularity, not an information limit: the oracle estimator (ability fit on the non-memorized items) still fires there. We report both estimators so the deployable envelope and the estimator-independent boundary are distinguished.

Uniform sandbagging mirrors this boundary: an across-the-board underperformer is re-estimated at a lower $\hat{\theta}$, and in the near-total limit no per-item anomaly survives on Z^- . Near-total uniform manipulation in either direction is absorbed into the ability estimate, while selective aberrance (the deployment-relevant case) remains detectable on both tails. The corner is also the least useful to an adversary: a uniform sandbagger surrenders capability everywhere and cannot hide one dangerous skill, which is the selective case the detector catches. This is therefore a single fundamental boundary at the near-total-uniform corner, and it means the method complements rather than replaces score-level and distributional detectors.

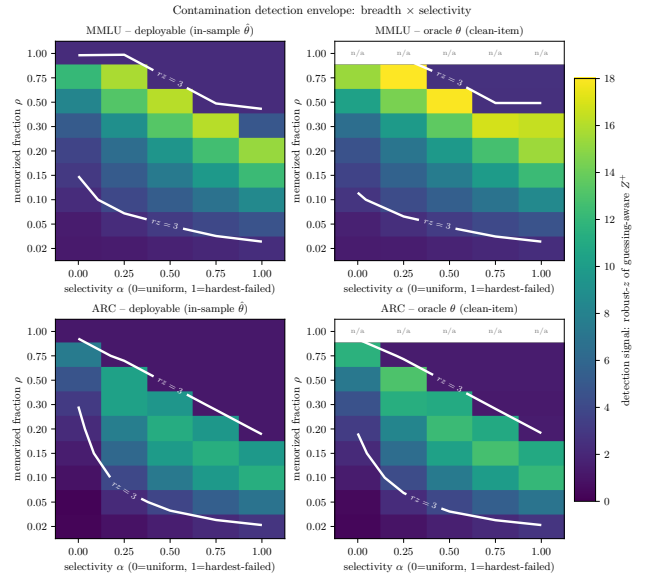


Figure 4: Detection envelope over memorization breadth ρ (rows) and selectivity α (columns); 0 = uniform-random, 1 = hardest-failed-first, for the deployable in-sample and oracle ability estimators on two multiple-choice benchmarks. Color is the guessing-aware Z^+ robust- z ; white contours mark the $z_{\text{rob}} = 3$ reporting threshold defined in “Empirical Null and Robust z ” and used throughout. The detector fires across most of the plane, including uniform partial memorization (left column), and is blind only in the near-total ($\rho \rightarrow 1$) corner. The interior in-sample collapse at high ρ is an ability-estimator artifact (the oracle panel still fires there), not an information boundary.

Conclusion

Benchmark contamination leaves a signature in the pattern of a model’s answers, not only in its aggregate score, and that signature is recoverable from the published leaderboard response matrix alone. Person-fit statistics detect injected contamination with a dose-response and distinguish it from benign fine-tuning; the signal is semantic rather than verbatim, surviving paraphrase; and a signed decomposition yields initial, small-sample evidence that contamination and sandbagging separate on a single axis. The method is a triage signal, not a standalone auditor: it is blind in the near-total-memorization corner, where inflated ability absorbs the leak. This operational envelope is mapped rather than assumed. As an access-minimal, psychometrically grounded complement to weight- and log-probability-based detectors, it extends contamination detection to the many models for which no deeper access exists.

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